Cost Model

Project 3: Governed Deployment Pattern (cross-cloud). Azure and AWS.

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Governance is usually treated as pure overhead. It is not. This pattern adds logging, monitoring, and safety cost, but the model-routing control removes inference cost, and at real volume the routing savings can exceed the governance overhead. A well-designed governed system can be cheaper than the same system run ungoverned on a single frontier model. This model records each governance component, its pricing basis, and which direction it moves cost.

Governance component	Pricing basis	Direction and magnitude
Audit logging	Azure Analytics Logs $2.30/GB (Basic $0.50/GB), 5 GB free monthly; AWS CloudWatch Logs $0.50/GB	Overhead, small: a decision record is a few KB
Content safety and PII guardrail	AWS Bedrock Guardrails (PII and content filters, per 1,000 text units); Azure AI Content Safety, comparable per request	Overhead, roughly a tenth of a cent per request
Monitoring and evaluation	Metrics plus periodic eval runs (model calls)	Overhead, small: periodic, not per request
Model routing	Bedrock Intelligent Prompt Routing (no routing surcharge, you pay the routed model); Azure model tiering	Savings, roughly 30 to 65% of inference
Human-in-the-loop gate	Human reviewer time	Operational cost, not cloud spend

Assumptions: pay-as-you-go rates, single-user test volume for the built proof, and decision records logged as small text entries rather than verbose traces. Azure Content Safety and the guardrail rates are directional, since exact per-record pricing varies by region and configuration.

At test volume the whole governed pattern still costs cents. A logged decision is a few kilobytes, so even thousands of decisions stay inside Azure's 5 GB free logging tier and AWS's low per-GB rate. The overhead only becomes material at enterprise volume, and that is exactly where routing starts paying for it.

The cost levers, both sides

On overhead, the lever is log tiering. Azure charges $2.30/GB for Analytics Logs but only $0.50/GB for Basic and $0.05/GB for Auxiliary, so keeping decision records in the queryable Analytics tier while pushing verbose debug traces to Basic or Auxiliary cuts logging cost by most of an order of magnitude. Naming that you would tier logs rather than dumping everything into the expensive tier is a senior cost move most people miss.

On savings, the lever is routing by query complexity. Sending simple requests to a cheaper model and only hard ones to the frontier model removes a large share of inference cost, and at enterprise volume that saving typically outweighs the logging, safety, and monitoring overhead. The one genuinely new cost the pattern adds is the human-in-the-loop reviewer, but that is human time, not cloud spend, and a high-risk deployment requires it regardless.

Production cost shape

Across the three projects the cost shapes differ: Project 1 was floor-dominated by a fixed search unit, Project 2 was pure usage with no floor, and Project 3 adds a governance layer whose net cost can be negative at scale, because routing savings exceed governance overhead. Three builds with three distinct cost shapes, fixed-floor, usage-metered, and governance-that-can-pay-for-itself, is the cost-structure fluency an enterprise architect is hired for.